ON CUTOFF VIA RIGIDITY FOR HIGH-DIMENSIONAL CURVED DIFFUSIONS

DJALIL CHAFAÏ AND MAX FATHI

ABSTRACT. We consider overdamped Langevin diffusions in Euclidean space, with curvature equal to the spectral gap. This includes the Ornstein–Uhlenbeck process as well as non-Gaussian and non-product extensions with convex interaction, such as the Dyson process from random matrix theory. We show that a cutoff phenomenon or abrupt convergence to equilibrium occurs in high dimension, at a critical time equal to the logarithm of the dimension divided by twice the spectral gap. This cutoff holds for Wasserstein distance, total variation, relative entropy, and Fisher information. A key observation is a relation to a spectral rigidity, linked to the presence of a Gaussian factor. A novelty is the extensive usage of functional inequalities, even for short-time regularization, and the reduction to Wasserstein. The proofs are short and conceptual. Since the product condition is satisfied, an Lp cutoff holds for all p. We moreover discuss a natural extension to Riemannian manifolds, a link with logarithmic gradient estimates in short-time for the heat kernel, and ask about stability by perturbation. Finally, beyond rigidity but still for diffusions, a discussion around the recent progress on the product condition for non-negatively curved diffusions leads us to introduce a new curvature product condition.

CONTENTS

1. Introduction and main results 2
1.1. Diffusions on Euclidean spaces, with convex potential 2
1.2. Cutoff for high-dimensional curved diffusions 2
1.3. Product condition and L^p cutoff at the mixing time 5
1.4. Positively curved diffusions, curvature product condition, and cutoff 6
1.5. Rigidity and Gaussian factorization for curved diffusions 6
1.6. Riemannian manifolds 7
1.7. About stability 9
Acknowledgments 9
2. Further comments 9
2.1. Fisher information and the heat kernel 9
2.2. Optimal constants 10
2.3. Normalization 10
2.4. Temperature 10
3. Proofs 11
3.1. Preliminaries on rigidity 11
3.2. Proof of Theorem 1.1 11
3.3. Proof of Corollary 1.3 12
3.4. Proof of Corollary 1.4 13
3.5. Proof of Theorem 1.5 14
3.6. Proof of Corollary 1.6 14
3.7. Proof of Theorem 1.7 15
3.8. Proof of Theorem 1.9 16
References 17

Date: Autumn 2024; Revised Spring 2025; Accepted Summer 2025; Published Autumn 2025; Revised Winter 2026. Postpublication version for *Comptes Rendus. Mathématique 2025, Vol. 363, p. 1103-1121* DOI: 10.5802/crmath.776.

2020 *Mathematics Subject Classification.* Primary 60J60; Secondary 58J65; 47D07; 53C21; 58J50.

Key words and phrases. Markov diffusion process; Curvature-dimension inequality; Spectral gap; Ergodicity and Cutoff; Wasserstein distance; Relative entropy; Total variation; Fisher information.

1. Introduction and main results

1.1. **Diffusions on Euclidean spaces, with convex potential.** Let $(X_t)_{t\geq 0}$ be the Markov diffusion process solving the stochastic differential equation (SDE)

$$\mathrm{d}X_t = -\nabla V(X_t)\mathrm{d}t + \sqrt{2}\mathrm{d}B_t, \quad X_0 = x_0 \in \mathbb{R}^d, \tag{1.1}$$

where $(B_t)_{t\geq 0}$ is a standard Brownian motion in $\mathbb{R}^d$, $V : \mathbb{R}^d \to \mathbb{R}$ is strictly convex and $\mathscr{C}^2$ with $\lim_{|x|\to\infty} V(x) = +\infty$, and $|\cdot|$ is the Euclidean norm of $\mathbb{R}^d$. In Statistical Physics, this drift-diffusion is also known as an overdamped Langevin process with potential V. By adding a constant to V, we can assume without loss of generality that $\mu = \mathrm{e}^{-V}$ namely

$$\mathrm{d}\mu(x) = \mathrm{e}^{-V(x)}\mathrm{d}x \tag{1.2}$$

is a probability measure. It is the unique invariant law of the process, and it is moreover reversible. The associated infinitesimal generator is the linear differential operator

$$\mathscr{L} = \Delta - \nabla V \cdot \nabla \tag{1.3}$$

acting on smooth functions. It is symmetric in $L^2(\mu)$, and its kernel is the set of constant functions. Moreover, its spectrum is included in $(-\infty, -\lambda_1] \cup \{0\}$, for some $\lambda_1 > 0$ called the spectral gap of $\mathscr{L}$. The Ornstein–Uhlenbeck (OU) process is obtained when $V(x) = \frac{\rho}{2}|x|^2$, $\rho > 0$, for which $\lambda_1 = \rho$ while $\mathrm{Hess}(V)(x) = \rho \mathrm{I}_d$ for all $x \in \mathbb{R}^d$.

If V is ρ-convex for some $\rho > 0$, namely if $V - \frac{\rho}{2}|\cdot|^2$ is convex, then the spectral gap is an eigenvalue of $-\mathscr{L}$, since the spectrum is discrete, see [Gigli et al., 2015, Prop. 6.7].

1.2. **Cutoff for high-dimensional curved diffusions.** Let us denote by W_2 the L^2 Wasserstein (or Monge–Kantorovich) coupling distance between probability measures on the same metric space with finite second moment, namely

$$\mathrm{W}_2(\nu, \mu) = \inf_{(X,Y)} \sqrt{\mathbb{E}(d(X,Y)^2)} \tag{1.4}$$

where the infimum runs over all couples (X, Y) of random variables with $X \sim \nu$ and $Y \sim \mu$, see Santambrogio [2015]. We also write $\mathrm{W}_2(Z, \mu) = \mathrm{W}_2(\mathrm{Law}(Z), \mu)$. We take $d(x, y) = |x - y|$ on $\mathbb{R}^k$.

Theorem 1.1 (Main Wasserstein estimate)**.** *Let $(X_t)_{t\geq 0}$ be the process* (1.1)*, with potential V, spectral gap λ_1, and invariant law $\mu = \mathrm{e}^{-V}$. If, for all $x \in \mathbb{R}^d$, and as quadratic forms,*

$$\mathrm{Hess}(V)(x) \geq \lambda_1 \mathrm{I}_d, \tag{1.5}$$

then for all non-empty set of initial conditions $S \subset \mathbb{R}^d$,

$$\frac{\mathrm{e}^{-2\lambda_1 t}}{\lambda_1} \sup_{x_0\in S} \Lambda(x_0) + \frac{k_1\big(1 - \sqrt{1 - \mathrm{e}^{-2\lambda_1 t}}\big)^2}{\lambda_1} \leq \sup_{x_0\in S} \mathrm{W}_2(X_t, \mu)^2 \leq \mathrm{e}^{-2\lambda_1 t} \sup_{x_0\in S} \int |x - x_0|^2 \mathrm{d}\mu(x) \tag{1.6}$$

with, denoting E_1 the eigenspace of $-\mathscr{L}$ associated to λ_1 and $k_1 = \dim(E_1)$,

$$\Lambda(x_0) = \sup_{(f_1,\dots,f_{k_1})} \sum_{i=1}^{k_1} |f_i(x_0)|^2, \tag{1.7}$$

where the supremum runs over the set of orthonormal bases of E_1.
Moreover, if μ is centered and $S = \{x \in \mathbb{R}^d : |x| \leq R\}$, then $\sup_{x_0\in S} \Lambda(x_0)$ can be replaced by $\lambda_1 R^2$.

Remark 1.2 (k_1 and $\Lambda(x_0)$)**.** *We always have $k_1 \leq d$, see [Barthe and Klartag, 2020, Lem. 14]. The factor involving k_1 could be dropped in the lower bound, since it is nonnegative. This will be good enough for the proof of Corollary 1.3 below, but there are situations, such as very small sets of initial data, where the extra factor is useful to get sharp bounds.*
By elliptic regularity, the elements of E_1 are $\mathscr{C}^2$ and in particular continuous, see [Evans, 2010, Ch. 6]. Moreover, since E_1 is finite dimensional, the linear map $f \in E_1 \mapsto \ell(f) = f(x_0)$ is continuous, and by the Riesz representation theorem, there exists an element $h_{x_0} \in E_1$ such that

$\ell = \langle \cdot, h_{x_0} \rangle$, and then the Parseval identity gives $\sum_{i=1}^{k_1} |f_i(x_0)|^2 = \sum_{i=1}^{k_1} |\langle f_i, h_{x_0}\rangle|^2 = \|h_{x_0}\|^2$ for all orthonormal basis $f_1, \dots, f_{k_1}$ of E_1. Therefore the supremum in the definition of $\Lambda(x_0)$ is useless.

Corollary 1.3 (Wasserstein cutoff)**.** *Let $(X_t^{(d)})$, $V^{(d)}$, $\lambda_1^{(d)}$, $\mu^{(d)}$ be as in Theorem 1.1, satisfying* (1.5) *for any dimension d. Let $m^{(d)}$ be the mean of $\mu^{(d)}$, and assume that*

$$\liminf_{d\to\infty} \lambda_1^{(d)} > 0. \tag{1.8}$$

Then a cutoff phenomenon occurs at critical time

$$t_* = \frac{\log(d)}{2\lambda_1^{(d)}}, \tag{1.9}$$

namely for all $\varepsilon \in (0,1)$,

$$\lim_{d\to\infty} \sup_{x_0^{(d)} \in S^{(d)}} \mathrm{W}_2(X_{t_d}^{(d)}, \mu^{(d)}) = \begin{cases} +\infty & \text{if } t_d = (1-\varepsilon)\, t_* \\ 0 & \text{if } t_d = (1+\varepsilon)\, t_* \end{cases} \tag{1.10}$$

where the set of initial conditions is a ball of the following form

$$S^{(d)} = \{x \in \mathbb{R}^d : |x - m^{(d)}| \le c\sqrt{d}\}, \quad \text{for an arbitrary constant } c > 0. \tag{1.11}$$

The assumption $c > 0$ is essential to get a cutoff at critical time $\log(d)/(2\lambda_1^{(d)})$. Indeed, in the case of the d-dimensional Ornstein–Uhlenbeck process with $S^{(d)} = \{m^{(d)}\}$ and spectral gap $\lambda_1^{(d)} = \rho > 0$ independent of d, one has a cutoff at critical time $\log(d)/(4\lambda_1^{(d)})$, and this is an effect of the second part of the left hand side of (1.6).

For probability measures μ and ν on the same space, we denote by

$$\mathrm{d_{TV}}(\mu, \nu) = \sup_A |\mu(A) - \nu(A)| \in [0,1] \tag{1.12}$$

their total variation distance, and by

$$\mathrm{H}(\nu \mid \mu) = \int \frac{\mathrm{d}\nu}{\mathrm{d}\mu} \log\Big(\frac{\mathrm{d}\nu}{\mathrm{d}\mu}\Big) \mathrm{d}\mu = \int \log\Big(\frac{\mathrm{d}\nu}{\mathrm{d}\mu}\Big) \mathrm{d}\nu \in [0, +\infty] \tag{1.13}$$

the relative entropy or Kullback–Leibler divergence of ν with respect to μ, with convention $\mathrm{H}(\nu \mid \mu) = +\infty$ if ν is not absolutely continuous with respect to μ. The Fisher information of ν with respect to μ is

$$\mathrm{I}(\nu \mid \mu) = \int \frac{|\nabla \frac{\mathrm{d}\nu}{\mathrm{d}\mu}|^2}{\frac{\mathrm{d}\nu}{\mathrm{d}\mu}} \mathrm{d}\mu = \int \Big|\nabla \log\Big(\frac{\mathrm{d}\nu}{\mathrm{d}\mu}\Big)\Big|^2 \mathrm{d}\nu \in [0, +\infty], \tag{1.14}$$

with the convention $\mathrm{I}(\nu \mid \mu) = +\infty$ if ν is not absolutely continuous with respect to μ. By an abuse of notation, we take the freedom of replacing ν by $X \sim \nu$ in these expressions.

Corollary 1.4 (I, H, and TV cutoffs)**.** *Under the setting of Corollary 1.3, and if additionally*

$$\lambda_1^{(d)} = o_{d\to\infty}(\log(d)) \tag{1.15}$$

then, for all $\varepsilon \in (0,1)$,

$$\lim_{d\to\infty} \sup_{x_0^{(d)} \in S^{(d)}} \mathrm{I}(X_{t_d}^{(d)} \mid \mu^{(d)}) = \begin{cases} +\infty & \text{if } t_d = (1-\varepsilon)\, t_* \\ 0 & \text{if } t_d = (1+\varepsilon)\, t_* \end{cases} \tag{1.16}$$

$$\lim_{d\to\infty} \sup_{x_0^{(d)} \in S^{(d)}} \mathrm{H}(X_{t_d}^{(d)} \mid \mu^{(d)}) = \begin{cases} +\infty & \text{if } t_d = (1-\varepsilon)\, t_* \\ 0 & \text{if } t_d = (1+\varepsilon)\, t_* \end{cases} \tag{1.17}$$

$$\lim_{d\to\infty} \sup_{x_0^{(d)} \in S^{(d)}} \mathrm{d_{TV}}(X_{t_d}^{(d)}, \mu^{(d)}) = \begin{cases} 1 & \text{if } t_d = (1-\varepsilon)\, t_* \\ 0 & \text{if } t_d = (1+\varepsilon)\, t_* \end{cases}. \tag{1.18}$$

We emphasize that in contrast with what is done for instance by Boursier et al. [2023] and Salez [2025], we obtain the I, H, and TV cutoffs from the Wasserstein cutoff. Moreover, they occur at exactly the same critical time, due to the choice of initial condition we make. In comparison, Salez [2025] works under more general curvature assumptions, and uses Fisher information and entropy to derive a differential inequality controlling entropy beyond the mixing time, and then proves cutoff in total variation distance using a reverse Pinsker inequality. All these approaches are crucially based on functional inequalities.

Condition (1.5) states that the process has curvature at least equal to the spectral gap. It turns out that it is the best possible lower bound on the curvature, as we explain later on in relation with a notion of spectral rigidity. Condition (1.5) is satisfied by the OU process with $V = \frac{\rho}{2}|\cdot|^2$, and in this case, we have $\lambda_1 = \rho$, $k_1 = d$, and $f_i(x) = \sqrt{\rho}x_i$. An important class of non-Gaussian and non-product examples beyond pure OU is

$$V(x) = \frac{\rho}{2}|x|^2 + W(x), \quad x \in \mathbb{R}^d, \tag{1.19}$$

where $\rho > 0$ and $W : \mathbb{R}^d \to \mathbb{R}$ is convex and translation invariant in the direction $(1,\dots,1) \in \mathbb{R}^d$, namely for all $u \in \mathbb{R}$ and all $x \in \mathbb{R}^d$,

$$W(x + u(1,\dots,1)) = W(x). \tag{1.20}$$

This is the case for example when for some convex even function $h : \mathbb{R} \to \mathbb{R}$,

$$W(x) = \sum_{i<j} h(x_i - x_j), \quad x \in \mathbb{R}^d. \tag{1.21}$$

If π and $\pi^\perp$ are the orthogonal projections on $\mathbb{R}(1,\dots,1)$ and its orthogonal, respectively, then $|x|^2 = |\pi(x)|^2 + |\pi^\perp(x)|^2$, while the translation invariance of W in the direction $(1/\sqrt{d},\dots,1/\sqrt{d})$ gives $W(x) = W(\pi(x) + \pi^\perp(x)) = W(\pi^\perp(x))$, therefore

$$\mathrm{e}^{-V(x)} = \mathrm{e}^{-\frac{\rho}{2}|\pi(x)|^2}\mathrm{e}^{-(W(\pi^\perp(x)) + \frac{\rho}{2}|\pi^\perp(x)|^2)} \tag{1.22}$$

which means that μ is, up to a rotation, a product measure, and splits into a one-dimensional Gaussian factor $\mathcal{N}(0, \frac{1}{\rho})$ and a log-concave factor with a ρ-convex potential.

Theorem 1.5 (Boltzmann–Gibbs probability measure with convex interaction)**.** *Let us consider the Langevin process* (1.1) *with V as in* (1.19) *with $\rho > 0$ and with W convex and translation invariant in the direction* $(1,\dots,1)$*. Then the following properties hold true:*

(i) $\lambda_1 = \rho$ and the symmetric Hermite polynomial $x_1 + \cdots + x_d$ belongs to E_1.
(ii) V is ρ-convex and μ has a Gaussian factor $\mathcal{N}(0, \frac{1}{\rho})$ in the direction $(1,\dots,1)$*.*
(iii) The curvature condition (1.5) *is satisfied.*

This approach can be adapted to the special degenerate case of the Dyson–OU or DOU process studied in Chafaï and Lehec [2020], Boursier et al. [2023], when

$$h(x) = \begin{cases} -\beta\log(x) & \text{if } x > 0 \\ +\infty & \text{if } x \le 0 \end{cases}, \quad \text{for an arbitrary constant } \beta \ge 0, \tag{1.23}$$

the degeneracy being equivalent to define the Dyson–OU process on the convex domain $\{x \in \mathbb{R}^d : x_1 > \cdots > x_d\}$ instead of on the whole space $\mathbb{R}^d$, to exploit convexity. In this case, the symmetric Hermite polynomial $x_1 + \cdots + x_d$ is an eigenfunction associated to the spectral gap, the tip of an iceberg of integrability, as observed in Lassalle [1991].

Corollary 1.6 (Cutoff for Langevin with convex interactions)**.** *Let $(X_t^{(d)})$, $V^{(d)}$, $\mu^{(d)}$ be as in Theorem 1.5, for any dimension d, and for a fixed $\rho > 0$. Then there is cutoff at critical time*

$$t_* = \frac{\log(d)}{2\rho} \tag{1.24}$$

in the sense that (1.10), (1.16), (1.17)*, and* (1.18) *occur when $S^{(d)}$ can be either $B(m^{(d)}, c\sqrt{d})$ or $m^{(d)} + [-c,c]^d$ for an arbitrary fixed constant $c > 0$ with $m^{(d)}$ being the mean of $\mu^{(d)}$.*

To summarize, for rigid curved diffusions with $\liminf_{d\to\infty} \lambda_1^{(d)} > 0$ and $\lambda_1^{(d)} = o_{d\to\infty}(\log d)$, the mixing time in Wasserstein distance, relative entropy, and total variation distance is always $\frac{\log(d)}{2\lambda_1}$, and a cutoff occurs at the mixing time in high dimension, for an explicit class of initial conditions. This covers non-Gaussian non-product examples, beyond the (D)OU case.

1.3. **Product condition and L^p cutoff at the mixing time.** It seems that the cutoff phenomenon for diffusions was first explored by Saloff-Coste [1994], notably for Brownian motion on compact Lie groups, using the L^2 decomposition and functional inequalities. This was further explored for manifolds with special symmetries in relation with representation theory by Méliot [2014]. The cutoff for diffusions on non-compact spaces such as the Dyson–OU process, is considered by Boursier et al. [2023], for various distances and divergences, in relation with integrability. The OU process is a special Dyson–OU process, and is also a special Gaussian ergodic Markov process with independent components, a tensorized or product situation for which the cutoff was studied earlier notably by Lachaud [2005], Barrera et al. [2006], and by Barrera [2018]. The role of the eigenfunctions associated to the spectral gap for lower bounds is an old observation that dates back to Persi Diaconis and David Wilson, see for instance Saloff-Coste [1994] and Boursier et al. [2023] and references therein for the case of diffusions.

It was shown by Chen and Saloff-Coste [2008] that for ergodic Markov processes on arbitrary state spaces, cutoff occurs at the mixing time, in L^p distance, $p > 1$, provided that the product of the spectral gap and the mixing time tends to infinity. This *product condition* was proposed by Yuval Peres. The method relies on a reduction to the Euclidean case $p = 2$ by interpolation. It does not provide the mixing time. Very recently, Salez [2025] extended this theorem to $p = 1$ (total variation distance), in the case of non-negatively curved diffusions. The method relies crucially on a two-sided comparison between total variation and relative entropy, namely Pinsker and reverse Pinsker inequalities, in other words functional inequalities. It is a variant of the method developed by Salez [2024] for finite state spaces, with a crucial specificity. Namely, it is shown that in the continuous setting a varentropy functional is controlled by the derivative of the entropy via a local Poincaré inequality, which arises as a consequence of non-negative curvature. In the discrete setting, chain rule issues are an obstruction, and current known statements have some extra restrictions, see Pedrotti and Salez [2025].

Under the hypotheses of Corollary 1.4, including $\liminf_{d\to\infty} \lambda_1^{(d)} > 0$ and $\lambda_1^{(d)} = o_{d\to\infty}(\log d)$, rigidity implies the product condition, and the mixing time in total variation goes to infinity by comparing with the projection of the process on an (affine) eigenfunction associated to the spectral gap. This allows to use [Chen and Saloff-Coste, 2008, Th. 3.3] and [Salez, 2025, Cor. 1] to get cutoff for L^p distance, $p \geq 1$, namely, denoting $\mu_t^{(d)}$ the law of $X_t^{(d)}$, this writes

$$\lim_{d\to\infty} \sup_{x_0^{(d)} \in S^{(d)}} \left\| \frac{\mathrm{d}\mu_{t_d}^{(d)}}{\mathrm{d}\mu^{(d)}} - 1 \right\|_{\mathrm{L}^p(\mu^{(d)})} = \begin{cases} \max & \text{if } t_d = (1-\varepsilon)\, t_* \\ 0 & \text{if } t_d = (1+\varepsilon)\, t_* \end{cases} \tag{1.25}$$

where max = 2 if $p = 1$ and max = $+\infty$ if $p > 1$, see Chafaï [2024] for a pedagogical presentation. We emphasize that Corollaries 1.3, 1.4, and 1.6 give the mixing time as well as the cutoff.

The product condition can be seen as weak rigidity, connecting trend to equilibrium and spectral gap. It is well known that in the discrete setting, the product condition does not always imply cutoff, see for instance Chen and Saloff-Coste [2008] for detailed counter examples. Discrete state space processes can be more subtle than random walks on locally Euclidean graphs, which are the discrete analogue of diffusions on manifolds. In particular they can allow complete graphs moves producing quick convergence to equilibrium. For diffusions, the analogue of this type of degeneracy would be a non-local term.

In another direction, for e^{-V} log-concave on $\mathbb{R}^d$, the Kannan, Lovász and Simonovitz (KLS) conjecture states that the inverse spectral gap λ_1^{-1} of $\Delta - \nabla V \cdot \nabla$ is up to universal multiplicative

constants equal to the spectral radius $r(K)$ of the covariance matrix

$$K_{ij} = \int x_i x_j \mathrm{d}\mu(x) - m_i m_j \quad \text{where} \quad m_i = \int x_i \mathrm{d}\mu(x).$$

The best available upper bound is $\lambda_1^{-1} \leq c_d r(K)$ with c_d logarithmic in d, see Klartag [2023]. The reverse bound $r(K) \leq \lambda_1^{-1}$ is always available.

1.4. **Positively curved diffusions, curvature product condition, and cutoff.** Recall that the recent result [Salez, 2025, Cor. 1] states that for non-negatively curved diffusions, and for total variation distance, the product condition implies cutoff at the mixing time (a direct corollary of a theorem bounding the mixing time window). By combining the approaches used in the proof of [Salez, 2025, Cor. 1], the proof of our Corollary 1.4, and the proof of [Chen and Saloff-Coste, 2008, Th. 3.3], we arrive at the following theorem. We state it in the restricted Euclidean setting for simplicity, but it holds in the Riemannian setting, since it is the curvature bound and the diffusion property that matter.

Theorem 1.7 (Cutoff for positively curved diffusions with curvature product condition)**.** *Let $(X_t^{(d)})$ be a sequence of diffusion processes on $\mathbb{R}^d$ of the form* (1.1)*, with convex potential $V^{(d)}$, initial condition $x_0^{(d)}$, and invariant measure $\mu^{(d)}$. Let $S^{(d)} \subset \mathbb{R}^d$ be an arbitrary non-empty set of initial conditions. Let $\eta \in (0,1)$ be an arbitrary fixed threshold which does not depend on d. Suppose that there exists positive constants $\kappa^{(d)}$ such that* $\mathrm{Hess}(V^{(d)}) \geq \kappa^{(d)} I_d$ *and that the following curvature product condition holds:*

$$\lim_{d\to\infty} \kappa^{(d)} t_0^{(d)} = +\infty \quad \textit{where} \quad t_0^{(d)} := \inf\{t \geq 0 : \sup_{x_0^{(d)} \in S^{(d)}} \mathrm{d_{TV}}(X_t^{(d)}, \mu^{(d)}) \leq \eta\}.$$

Assume also that $t_0^{(d)}$ is finite for all sufficiently large d. Then there is cutoff at the critical time $t_0^{(d)}$ in the sense that for all $\varepsilon \in (0,1)$,

$$\lim_{d\to\infty} \sup_{x_0^{(d)} \in S^{(d)}} \mathrm{d_{TV}}(X_{t_d}^{(d)}, \mu^{(d)}) = \begin{cases} 1 & \text{if } t_d = (1-\varepsilon) t_0^{(d)} \\ 0 & \text{if } t_d = (1+\varepsilon) t_0^{(d)} \end{cases} \tag{1.26}$$

$$\lim_{d\to\infty} \sup_{x_0^{(d)} \in S^{(d)}} \mathrm{H}(X_{t_d}^{(d)} \mid \mu^{(d)}) = \begin{cases} +\infty & \text{if } t_d = (1-\varepsilon) t_0^{(d)} \\ 0 & \text{if } t_d = (1+\varepsilon) t_0^{(d)} \end{cases} \tag{1.27}$$

and, under the stronger assumptions $\underline{\lim}_{d\to\infty} \kappa^{(d)} > 0$ and $\lim_{d\to\infty} t_0^{(d)} = +\infty$, we also have

$$\lim_{d\to\infty} \sup_{x_0^{(d)} \in S^{(d)}} \mathrm{I}(X_{t_d}^{(d)} \mid \mu^{(d)}) = \begin{cases} +\infty & \text{if } t_d = (1-\varepsilon) t_0^{(d)} \\ 0 & \text{if } t_d = (1+\varepsilon) t_0^{(d)} \end{cases} \tag{1.28}$$

$$\lim_{d\to\infty} \sup_{x_0^{(d)} \in S^{(d)}} \mathrm{W}_2(X_{t_d}^{(d)}, \mu^{(d)}) = \begin{cases} +\infty & \text{if } t_d = (1-\varepsilon) t_0^{(d)} \\ 0 & \text{if } t_d = (1+\varepsilon) t_0^{(d)} \end{cases}. \tag{1.29}$$

1.5. **Rigidity and Gaussian factorization for curved diffusions.** It turns out that the presence of a Gaussian factor, as well as the λ_1 uniform lower bound on the Hessian of the potential (1.5), are both equivalent to a notion of rigidity.

Let us start with the notion of Bakry–Émery curvature. For a Langevin process in the Euclidean space as in (1.1), for all $\rho > 0$, the following properties are equivalent:

(C1) $V - \frac{\rho}{2}|\cdot|^2$ is convex, in other words V is ρ-convex
(C2) $\mathrm{Hess}(V)(x) \geq \rho \mathrm{I}_d$ as quadratic forms, for all $x \in \mathbb{R}^d$
(C3) $\langle \mathrm{Hess}(V)\nabla f, \nabla f\rangle \geq \rho |\nabla f|^2$ for all smooth $f : \mathbb{R}^d \to \mathbb{R}$.
(C4) the curvature-dimension inequality $\mathrm{CD}(\rho,\infty)$ is satisfied, see Section 1.6.

Following Bakry et al. [2014], when these properties hold true, we say that the process has (Bakry–Émery) curvature bounded below by ρ. It follows then that

$$\lambda_1 \geq \rho. \tag{1.30}$$

To see it, we observe that from the Bochner formula $\nabla\mathscr{L} - \mathscr{L}\nabla = -\mathrm{Hess}(V)\nabla$, we get

$$\int \big(\langle \mathrm{Hess}(V)\nabla f, \nabla f\rangle - \rho|\nabla f|^2\big)\mathrm{d}\mu = \int \big(\langle \mathscr{L}\nabla f - \nabla\mathscr{L} f, \nabla f\rangle - \rho|\nabla f|^2\big)\mathrm{d}\mu, \tag{1.31}$$

while on the other hand, by integration by parts, we get[1]

$$-\int \langle \mathscr{L}\nabla f, \nabla f\rangle \mathrm{d}\mu = \sum_{i=1}^{d}\int |\nabla\partial_i f|^2\mathrm{d}\mu = \sum_{i,j=1}^{d}\int (\partial^2_{ij} f)^2\mathrm{d}\mu = \int \|\mathrm{Hess}(f)\|^2_{\mathrm{HS}}\mathrm{d}\mu, \tag{1.32}$$

therefore, by specializing to f such that $-\mathscr{L} f = \lambda_1 f$, we obtain

$$\int \big(\underbrace{\langle \mathrm{Hess}(V)\nabla f, \nabla f\rangle - \rho|\nabla f|^2}_{\geq 0 \text{ by (C3)}}\big)\mathrm{d}\mu = (\lambda_1 - \rho)\int |\nabla f|^2\mathrm{d}\mu - \int \|\mathrm{Hess}(f)\|^2_{\mathrm{HS}}\mathrm{d}\mu, \tag{1.33}$$

which gives (1.30)[2]. Equality in (1.30) is obviously achieved for the OU case $V = \frac{\rho}{2}|\cdot|^2$. More generally, beyond the OU case, we say that the process is *rigid* when the spectral gap matches the curvature lower bound: $\lambda_1 = \rho$. Rigidity can be reformulated, and it turns out that for all $\rho > 0$, the following items are equivalent:

- (R1) The process is rigid: it has curvature bounded below by ρ and $\lambda_1 = \rho$
- (R2) Up to a rotation and translation (change of coordinates), the probability measure $\mu = \mathrm{e}^{-V}$ is product, and splits into a one-dimensional Gaussian factor $\mathcal{N}(0, \frac{1}{\rho})$ and a second factor which is log-concave with a ρ-convex potential. In which case, we get $\lambda_1 = \rho$ and all the eigenfunctions associated to λ_1 are affine.

It is immediate to check that (R2) gives (R1). Conversely, if (R1) is satisfied, then for f such that $-\mathscr{L} f = \lambda_1 f$, we obtain that both sides of (1.33) are identically zero. This gives $\mathrm{Hess}(f) \equiv 0$, hence f is affine, and this also gives that the constant vector ∇f is in the kernel of the symmetric positive semidefinite matrix $\mathrm{Hess}(V) - \rho \mathrm{I}_d$. Moreover $\nabla f \neq 0$ since f is not constant, leading to (R2). We observe that by iterating the procedure, we get a Gaussian factor with same dimension as the eigenspace of λ_1.

The equivalence between the rigidity properties (R1) and (R2) appears in Cheng and Zhou [2017] in the traditional broader Riemannian setting of Bakry–Émery, discussed in Section 3.8. An alternative proof in the Euclidean case, based on optimal transport, was found by De Philippis and Figalli [2017]. These works were motivated by the more general problem of understanding manifolds optimizing a certain geometric quantity (namely, the spectral gap) under a curvature constraint. This type of question goes back to a theorem of Obata [1962], which states that positively curved smooth Riemannian manifolds with minimal spectral gap are isometric to spheres. Rigidity is typically used to reinforce statements in comparison geometry. See for example Bakry and Qian [2000] for applications of Gamma calculus to comparison geometry under curvature assumptions.

1.6. **Riemannian manifolds.** We now discuss Theorem 1.1 in a broader geometric context. A weighted Riemannian manifold is a triplet (M, g, μ), where (M, g) is a Riemannian manifold with metric tensor g, and μ is a measure on M. Throughout the Riemannian setting, we assume that (M, g) is smooth, connected, complete, and without boundary. Here we shall assume that μ is a probability measure, absolutely continuous with respect to the volume measure. We shall write $\mu = \mathrm{e}^{-V}$, so that V plays the same role of a potential as in the Euclidean setting. The

[1] We denote by $\|\cdot\|_{\mathrm{HS}}$ the Hilbert–Schmidt or trace or Frobenius matrix norm.

[2] We could also invoke the Brascamp–Lieb inequality $\int f^2\mathrm{d}\mu - (\int f\mathrm{d}\mu)^2 \leq \int \langle \mathrm{Hess}(V)^{-1}\nabla f, \nabla f\rangle \mathrm{d}\mu$, or equivalently the covariance representation of Hörmander–Helffer–Sjöstrand.

Markov process $(X_t)_{t\geq 0}$ we consider is the drift diffusion that combines a Brownian motion $(B_t)_{t\geq 0}$ on (M,g) and drift $-\nabla V$, given by the SDE

$$\mathrm{d}X_t = -\nabla V(X_t)\mathrm{d}t + \sqrt{2}\mathrm{d}B_t, \quad X_0 = x_0 \in M, \tag{1.34}$$

where ∇ is the gradient on (M,g). We refer to Bakry et al. [2014] and Hsu [2002] for more background. A key role is played by the Ricci curvature tensor of the manifold, which we shall denote by Ric. We refer to Ollivier [2013] for an introduction to curvature on Riemannian manifolds. Under the assumption that the Ricci curvature is bounded from below, and that V is geodesically semi-convex, then solutions to (1.34) exist for all times [Grigor'yan, 2009, Th. 11.8]. We shall not make use of (1.34), and only rely on the generator of the process

$$\mathscr{L} = \Delta - \nabla V \cdot \nabla \tag{1.35}$$

where Δ is the Laplace–Beltrami operator on (M,g). The weighted manifold (M,g,μ) has Bakry–Émery curvature bounded below by ρ when

$$\mathrm{Ric} + \mathrm{Hess}(V) \geq \rho g. \tag{1.36}$$

A celebrated result due to Bakry and Émery [1985], see also Bakry et al. [2014], states that this implies a lower bound on the spectral gap of $\mathscr{L}$ in the sense that

$$\lambda_1 \geq \rho. \tag{1.37}$$

When the manifold is unweighted, this is the dimension-free version of the famous spectral gap bound due to Lichnérowicz [1958]. Actually the Bakry–Émery approach implies stronger functional inequalities, such as a logarithmic Sobolev inequality, and a Gaussian isoperimetric inequality. Note that following for instance [Gigli et al., 2015, Prop. 6.7], when $\rho > 0$ the spectrum of $\mathscr{L}$ is discrete and the spectral gap λ_1 is always an eigenvalue of $-\mathscr{L}$. Moreover, the eigenfunctions associated to the spectral gap are continuous.

In the Bakry–Émery approach, the bound (1.36) is formulated in a more abstract way involving the Γ and Γ_2 functional quadratic forms, defined by

$$\Gamma(f,g) = \tfrac{1}{2}(\mathscr{L}(fg) - f\mathscr{L}g - g\mathscr{L}f) \tag{1.38}$$

$$\Gamma_2(f,g) = \tfrac{1}{2}(\mathscr{L}\Gamma(f,g) - \Gamma(f,\mathscr{L}g) - \Gamma(g,\mathscr{L}f)). \tag{1.39}$$

They are fully characterized by their diagonal $\Gamma(f) = \Gamma(f,f)$ and $\Gamma_2(f) = \Gamma_2(f,f)$, by polarization. In the case of the Langevin operator (1.35), they simply boil down to

$$\Gamma(f) = |\nabla f|^2 \quad \text{and} \quad \Gamma_2(f) = \mathrm{Ric}(\nabla f, \nabla f) + \|\mathrm{Hess}(f)\|_{\mathrm{HS}}^2 + \langle \mathrm{Hess}(V)\nabla f, \nabla f\rangle, \tag{1.40}$$

The formula for Γ_2 follows from the Bochner formula $\nabla\mathscr{L} = \mathscr{L}\nabla - \mathrm{Hess}(V)\nabla - \mathrm{Ric}(\nabla,\nabla)$. The Bakry–Émery curvature-dimension inequality $\mathrm{CD}(\rho,n)$, where $\rho \in \mathbb{R}$ is the curvature and $n \in \mathbb{R}$ is the dimension, writes, for all f in a rich enough class of functions,

$$\Gamma_2(f) \geq \rho\Gamma(f) + \frac{1}{n}(\mathscr{L}f)^2. \tag{1.41}$$

Now $\mathrm{CD}(\rho,\infty)$ is equivalent to the curvature lower bound (1.36). Moreover, in the Euclidean case, the Ricci tensor is zero and $\mathrm{CD}(\rho,\infty)$ is equivalent to say that V is ρ-convex.

In order to formulate a neat abstract analogue of Theorem 1.1, we shall use vector-valued functions whose coordinates are eigenfunctions, as defined in the following:

Definition 1.8 (Multi-eigenfunction)**.** *Let E_1 be the eigenspace of $-\mathscr{L}$ associated with the eigenvalue λ_1, in $L^2(\mu)$. Let $k_1 = \dim E_1$. We define a multi-eigenfunction as being a map $M \to \mathbb{R}^{k_1}$ whose coordinates are orthogonal elements of E_1 in $L^2(\mu)$. We denote by F_1 the set of all multi-eigenfunctions, and by $\mathbb{S}_{F_1}$ the set of multi-eigenfunctions whose coordinates are elements of the unit sphere of E_1 in $L^2(\mu)$, thus orthonormal.*

The link between this definition and Theorem 1.1 is that the component $\sum_{i=1}^{k_1} |f_i(x_0)|^2$ in the definition of Λ in Theorem 1.1 is exactly the ℓ^2 norm of $T(x_0)$ where T is some element of $\mathbb{S}_{F_1}$. The Riemannian analog of Theorem 1.1 is now the following.

Theorem 1.9 (Wasserstein estimate on weighted Riemannian manifold)**.** *Let $(X_t)_{t\geq 0}$ be the diffusion* (1.34) *on a smooth, connected, complete Riemannian manifold without boundary, and let $\mathscr{L}$ be its generator* (1.35)*. Assume that for some $\rho > 0$,*

$$\mathrm{CD}(\rho,\infty) \text{ is satisfied and } \lambda_1 = \rho. \tag{1.42}$$

Then for all non-empty set of possible initial conditions $S \subset M$, and all $t \geq 0$,

$$\frac{\mathrm{e}^{-2\rho t}}{\rho} \sup_{x_0\in S} \Lambda(x_0) + \frac{k_1\big(1-\sqrt{1-\mathrm{e}^{-2\rho t}}\big)^2}{\rho} \leq \sup_{x_0\in S} \mathrm{W}_2(X_t,\mu)^2 \leq \mathrm{e}^{-2\rho t} \sup_{x_0\in S} \int d_M(x_0,x)^2 \mathrm{d}\mu(x), \tag{1.43}$$

where d_M is the Riemannian distance on M, and where

$$\Lambda(x) = \sup_{T\in\mathbb{S}_{F_1}} |T(x)|^2. \tag{1.44}$$

Moreover, if $S = \{x \in M : d_M(x,m) \leq R\}$ with $\Lambda(m) = 0$ then we can replace $\sup_{x_0\in S} \Lambda(x_0)$ by ρR^2.

1.7. **About stability.** Our study is under the strong assumption $\lambda_1 = \rho$. For the d-dimensional unit sphere, we have $\rho = d-1$ and $\lambda_1 = d$, so the abstract theorem does not apply. Yet cutoff does occur, and moreover $\lambda_1/\rho \longrightarrow 1$. This naturally leads to asking

Question 1.10. *Is there a (useful) analog of Corollary 1.3 under the weaker assumption*

$$\frac{\lambda_1^{(d)}}{\rho^{(d)}} \xrightarrow[d\to\infty]{} 1\,? \tag{1.45}$$

Some results on positively curved manifolds when λ_1 is close to ρ were obtained by Courtade and Fathi [2020], Mai and Ohta [2021], Bertrand and Fathi [2022], Fathi et al. [2024]. However, the quantitative estimates on eigenfunctions seem to be too weak to easily generalize Theorem 1.9. The estimate on the Wasserstein distance strongly relies on the fact that the eigenfunction is Lipschitz, and the estimate is sharp because it is actually affine, namely that $\mathrm{Hess}(f) = 0$. When λ_1 is close to ρ, then $\mathrm{Hess}(f)$ is small, but for example the smallness estimates in Bertrand and Fathi [2022] are only in L^2 norm, so we do not actually control the Lipschitz norm of f. It is unclear if a stronger L^∞ estimate can be expected in general, so maybe some extra assumptions are needed.

Acknowledgments. This work has been (partially) supported by the Project CONVIVIALITY ANR-23-CE40-0003 of the French National Research Agency.

2. Further comments

This section gathers several remarks, mostly about the Euclidean case for simplicity.

2.1. **Fisher information and the heat kernel.** It is natural to try to directly upper bound the Fisher information by using the heat kernel, instead of using a comparison with relative entropy. Following Boursier et al. [2023], let us examine first the exactly solvable case of the OU process, for which the Mehler formula $\mu_t = \mathscr{N}(\mathrm{e}^{-\rho t}x_0, \frac{1-\mathrm{e}^{-2\rho t}}{\rho}\mathrm{I}_d)$ gives, for $t > 0$,

$$\nabla \log \frac{\mathrm{d}\mu_t}{\mathrm{d}\mu}(x) = -\rho \frac{x - \mathrm{e}^{-\rho t}x_0}{1-\mathrm{e}^{-2\rho t}} + \rho x = a_t x + b_t x_0 \tag{2.1}$$

where a_t and b_t do not depend on the dimension d, leading to

$$\mathrm{I}(\mu_t \mid \mu) = \rho d \frac{\mathrm{e}^{-4\rho t}}{1-\mathrm{e}^{-2\rho t}} + \rho^2 \mathrm{e}^{-2\rho t}|x_0|^2. \tag{2.2}$$

This gives the upper bound (3.32) as well as the lower bound (3.27). Beyond the OU process, denoting, for all $t > 0$, by p_{t,x_0} the density of μ_t with respect to the Lebesgue measure, which is known as the heat kernel of $\mathscr{L}$, we have, for all x and all $t > 0$,

$$\nabla \log \frac{\mathrm{d}\mu_t}{\mathrm{d}\mu}(x) = \nabla \log p_{t,x_0}(x) + \nabla V(x). \tag{2.3}$$

Inspired by the OU example, we could expect that under $\mathrm{CD}(\rho,\infty)$, for t small enough,

$$\left|\nabla \log p_{t,x_0}(x)\right|^2 \le a_t^2|x|^2 + b_t^2|x_0|^2, \tag{2.4}$$

which recalls the parabolic logarithmic Harnack and Li–Yau inequalities, leading to

$$\mathrm{I}(\mu_t \mid \mu) \le 2\int (a_t^2|x|^2 + |\nabla V(x)|^2)\mathrm{d}\mu_t(x) + 2b_t^2|x_0|^2. \tag{2.5}$$

Here a_t and b_t are constants with respect to x, that may depend on t but not on d and x_0. The explicit dependence on the dimension and initial condition is crucial here for our purposes. Unfortunately, we were not able to locate (2.4) in the literature on short time logarithmic gradient bounds for the heat kernel on non-compact manifolds with a lower curvature bound. Close works are Engoulatov [2006] where a constant is not controlled with respect to the dimension, and Chen et al. [2023] which is curvature free but is restricted to a compact subset. It would not be a surprise to assume, in addition to positive curvature, an additional condition regarding eigenfunctions associated to the spectral gap.

To summarize, the upper bound on $\mathrm{I}(\mu_t \mid \mu)$ follows from a logarithmic gradient bound for the heat kernel in short time, which expresses a regularization property of the evolution equation driven by $\mathscr{L}$ when it starts from a Dirac mass. Such an upper bound on the Fisher information I implies all the others. Indeed, it would give an upper bound on H by using the logarithmic Sobolev inequality, giving in turn an upper bound on W_2 by using the Talagrand inequality, as well as an upper bound on d_{TV} by using the Pinsker inequality.

An estimate on I is at the heart of the varentropy approach in Salez [2024]. In Salez [2025], the control of Fisher information is circumvented by a direct comparison with varentropy.

2.2. **Optimal constants.** An observation that dates back to Bakry and Émery [1985], see also Bakry et al. [2014], is that $\mathrm{CD}(\rho,\infty)$ implies a logarithmic Sobolev inequality (LSI) with constant $2/\rho$ as well as a Poincaré inequality (PI) of constant $1/\rho$ (directly as well as by linearization). Recall that the optimal Poincaré constant is precisely the inverse of the spectral gap $1/\lambda_1$. It follows that rigidity implies that the optimal LSI constant is twice the optimal PI constant, just like for the OU process. By the way, a famous alternative due to Oscar Rothaus states that in the compact setting, if the optimal LSI constant is not twice the optimal PI constant, then there exists an extremal function for LSI, see for instance [Rothaus, 1981, Th. at bottom of p. 107] as well as [Saloff-Coste, 1997, Th. 2.2.3 p. 333] in the lecture notes by Laurent Saloff-Coste for a discrete version. Note also that in the case (1.19), the function $x \mapsto \mathrm{e}^{\alpha(x_1+\cdots+x_d)}$ is extremal for LSI, as observed in Chafaï and Lehec [2020], while $x \mapsto x_1+\cdots+x_d$ is extremal for PI.

2.3. **Normalization.** For any real $\alpha > 0$ which may depend on d, the time-changed process $X^{(\alpha)} := (X_{\alpha t})_{t\ge 0}$ solves $\mathrm{d}X_t^{(\alpha)} = -\alpha\nabla V(X_t^{(\alpha)})\mathrm{d}t + \sqrt{2\alpha}\mathrm{d}B_t$ and has generator $\alpha\mathscr{L}$. The process X has cutoff at critical time t_* iff $X^{(\alpha)}$ has cutoff at critical time t_*/α. Such scaled processes play a role with respect to mean-field limits of interacting particle systems related to McKean–Vlasov semilinear PDE, see Boursier et al. [2023]. Note that the product condition is invariant under such a scaling.

2.4. **Temperature.** For any real $\sigma > 0$ independent of d, playing the role of the temperature, let us consider the Markov diffusion process $(Y_t^{(\sigma)})_{t\ge 0}$ solving the stochastic differential equation $\mathrm{d}Y_t^{(\sigma)} = -\nabla V(Y_t^{(\sigma)})\mathrm{d}t + \sqrt{2\sigma^2}\mathrm{d}B_t$. This is (1.1) when $\sigma = 1$. Its invariant law is $\mathrm{d}\mu_\sigma(x) = Z_\sigma^{-1}\mathrm{e}^{-V(x)/\sigma^2}\mathrm{d}x$, where $Z_\sigma = \int_{\mathbb{R}^d}\mathrm{e}^{-V(x)/\sigma^2}\mathrm{d}x$, while its generator is $\sigma^2\Delta - \nabla V\cdot\nabla$. The Bakry–Émery operators are

$$\Gamma(f) = \sigma^2|\nabla f|^2 \quad \text{and} \quad \Gamma_2(f) = \sigma^4\|\mathrm{Hess} f\|_{\mathrm{HS}}^2 + \sigma^2\langle \mathrm{Hess}(V)\nabla f, \nabla f\rangle. \tag{2.6}$$

Thus, for all $\rho > 0$, $\mathrm{CD}(\rho,\infty)$ of $\mathscr{L}$ is equivalent to $\mathrm{Hess}(V) \ge \rho \mathrm{I}_d$ as quadratic form, which is free of σ. In particular, in the rigid case $\rho = \lambda_1$, the cutoff critical time is free of σ. This can be also obtained from Section 2.3 by using $\alpha = \sigma^2$ and $\frac{1}{\sigma^2}V$ instead of V.

3. PROOFS

3.1. **Preliminaries on rigidity.** As already mentioned, our starting point is a Gaussian splitting or factorization theorem of Cheng and Zhou [2017], which in the Euclidean setting takes the form of Theorem 3.1 below. See also Section 1.5 and [Chafaï and Lehec, 2020, Rem. 3.2].

Theorem 3.1 (Gaussian factorization in the Euclidean space)**.** *Consider the operator* (1.3) *and assume that for some constant $\rho > 0$, V is ρ-convex and $\lambda_1 = \rho$. Then there is an orthonormal basis $(e_1, \dots, e_d)$ of $\mathbb{R}^d$ and a vector $m \in \mathbb{R}^d$ such that V is of the form*

$$V(x) = \frac{\rho}{2}((x_1 - m_1)^2 + \cdots + (x_{k_1} - m_{k_1})^2) + \widetilde{V}(x_{k_1+1}, \dots, x_d) \tag{3.1}$$

where k_1 is the dimension of the eigenspace E_1 of $-\mathscr{L}$ associated with λ_1, and $\widetilde{V}$ is ρ-convex on $\mathbb{R}^{d-k_1}$. Moreover, all eigenfunctions with eigenvalue λ_1 are affine, and only depend on the first k_1 coordinates in the above basis.

The vector m actually is the center of mass of the probability measure e^{-V}.

Remark 3.2 (Eigenfunctions structure)**.** *By Theorem 3.1, if $f \in E_1$, then f is affine, and since $\int |\nabla f|^2 \mathrm{d}\mu = -\int (\mathscr{L} f) f \mathrm{d}\mu = \lambda_1 \int f^2 \mathrm{d}\mu$, we get $f(x) = \langle a, p(x) \rangle_{\mathbb{R}^{k_1}} + b$ with p the projection on the k_1-dimensional Gaussian factor and $|a|^2 = \lambda_1 \|f\|^2_{L^2(\mu)}$.*

Another way of stating this result is that up to a rotation (the change of basis) and a translation by the vector m, the law e^{-V} is a product measure, with a centered Gaussian factor with variance $1/\rho$ on the first k_1 coordinates, and a log-concave factor with a ρ-convex potential on the last $d - k_1$ coordinates. Note that the result can only be true up to a rotation and translation, since the assumptions are stable by isometries of $\mathbb{R}^d$.

An alternative proof in the Euclidean setting, based on a rigidity property for regularity of solutions to the Monge–Ampère PDE for optimal transport maps, was given by De Philippis and Figalli [2017]. We shall discuss some elements of proof in the full Riemannian setting in Section 3.8.

Lemma 3.3 (OU process associated to an eigenfunction)**.** *Under the setting of Theorem 3.1 and Definition 1.8, for all $T \in \mathbb{S}_{F_1}$, the process $(T(X_t))_{t \geq 0}$ is a k_1-dimensional OU process scaled by a factor ρ, that is a process on $\mathbb{R}^{k_1}$ with generator $\rho\Delta - \rho x \cdot \nabla$.*

Proof. Let us show that $T(X)$ is a Markov process and compute its generator. If $\vec{v}$ is a vector-valued function, then $\vec{\mathscr{L}}\vec{v}$ is the vector obtained by applying $\mathscr{L}$ to each coordinate, and $\Gamma(\vec{v})$ the matrix whose coefficients are $\Gamma(\vec{v}_i, \vec{v}_j)$. Since the coordinates of T are orthogonal normalized eigenfunctions, $\vec{\mathscr{L}} T = -\lambda_1 T$ and $\Gamma(T) = \rho\,\mathrm{Id}$, as per Theorem 3.1 (affine with orthogonal constant gradients of norm $\sqrt{\rho}$). By the diffusion property, we have

$$\begin{aligned} \mathscr{L}(g \circ T) &= \nabla g \circ T \cdot \vec{\mathscr{L}} T + \langle \nabla^2 g \circ T, \Gamma(T) \rangle && (3.2)\\ &= -\lambda_1 \nabla g \circ T \cdot T + \rho \Delta g \circ T && (3.3)\\ &= -\rho \nabla g \circ T \cdot T + \rho \Delta g \circ T. && (3.4) \end{aligned}$$

This is a function of T, so $T(X)$ is a Markov process, and when viewing it as such it is indeed equal to ρ times the generator of an OU process with unit variance, applied to a function g. □

Hence, in the rigid case, the full process contains an OU subprocess, and hence cannot converge to equilibrium faster than it. This will yield the lower bound in Theorem 1.1.

3.2. **Proof of Theorem 1.1.** The upper bound is an immediate well-known consequence of the exponential convergence to equilibrium in Wasserstein distance under the $\mathrm{CD}(\rho = \lambda_1, \infty)$ condition, see for instance von Renesse and Sturm [2005]: for all X_0, Y_0 and $t \geq 0$,

$$\mathrm{W}_2(X_t, Y_t) \leq \mathrm{e}^{-\rho t} \mathrm{W}_2(X_0, Y_0). \tag{3.5}$$

The desired upper bound follows by taking $Y_0 \sim \mu$ and X_0 deterministic.

Let us now prove the lower bound. Consider a multi-eigenmap T, see Definition 1.8. Up to an isometry, it is of the form $T(x) = Ax+b$. Moreover, since the coordinates of T are orthogonal eigenfunctions, the rows of A are orthogonal vectors. If moreover we assume that $T \in \mathbb{S}_{F_1}$, then each row of A has norm $\sqrt{\rho} = \sqrt{\lambda_1}$. Hence we get

$$\mathrm{W}_2(T(X_t), T(Y_t)) \leq \sqrt{\lambda_1}\mathrm{W}_2(X_t, Y_t). \tag{3.6}$$

Applying Lemma 3.3, $Z_t := T(X_t)$ and $T(Y_t)$ are OU processes with speed accelerated by a factor ρ. Suppose now that Y_0 is distributed according to the equilibrium measure: $Y_0 \sim \mu$. Then $T(Y_t)$ follows a standard Gaussian law γ_{k_1} on $\mathbb{R}^{k_1}$, for all t. Thus using the formula $\mathrm{W}_2(\mathcal{N}(m_1,\Sigma_1),\mathcal{N}(m_2,\Sigma_2))^2 = |m_1-m_2|^2 + \mathrm{Tr}((\sqrt{\Sigma_1}-\sqrt{\Sigma_2})^2)$ when $\Sigma_1\Sigma_2 = \Sigma_2\Sigma_1$, the Mehler formula $Z_t \sim \mathcal{N}(T(x_0)\mathrm{e}^{-\rho t}, (1-\mathrm{e}^{-2\rho t})I_{k_1})$, and $\gamma_{k_1} = \mathcal{N}(0, I_{k_1})$, we find

$$\mathrm{W}_2(Z_t, \gamma_{k_1})^2 = \mathrm{e}^{-2\rho t}|T(x_0)|^2 + k_1\big(1-\sqrt{1-\mathrm{e}^{-2\rho t}}\big)^2. \tag{3.7}$$

Optimizing over x_0 and T concludes the proof.

We now consider the case where μ is centered and $S = B(0,R)$. For any $T \in \mathbb{S}_{F_1}$, writing $T = Ax+b$, since $\int T\mathrm{d}\mu = 0$ we see that $b = 0$. Therefore $T(0) = 0$. Moreover, since the rows of A are orthogonal and have norm $\sqrt{\lambda_1}$, we get $\sup_{x\in S}|Ax|^2 = \lambda_1 R^2$.

3.3. **Proof of Corollary 1.3.** Let us drop the (d) superscript. Without loss of generality, we can assume that $m = 0$ by translating V. We start by using the lower bound of Theorem 1.1 to prove the convergence to infinity when $t_d = (1-\epsilon)\,t_*$. Since we are in the centered setting and the set of initial conditions is a centered ball of radius $c\sqrt{d}$, the lower bound

$$\frac{\mathrm{e}^{-2\lambda_1 t}}{\lambda_1}\sup_{x_0\in S}\Lambda(x_0) \leq \sup_{x_0\in S}\mathrm{W}_2(X_t,\mu)^2 \tag{3.8}$$

while $\sup_{x_0\in S}\Lambda(x_0) = \lambda_1 c^2 d$ thanks to Theorem 1.1 since $S = B(0, c\sqrt{d})$ with $c > 0$. Evaluating at $t = t_d$ and neglecting k_1, we get

$$cd^{\epsilon/2} \leq \sup_{x_0\in S}\mathrm{W}_2(X_{t_d},\mu) \tag{3.9}$$

and letting d go to infinity concludes the proof of the lower bound. Let us now prove the case $t_d = (1+\epsilon)\,t_*$ via the upper bound in Theorem 1.1. Since μ is centered,

$$\int |x-x_0|^2\mathrm{d}\mu(x) = |x_0|^2 + \int |x|^2\mathrm{d}\mu(x). \tag{3.10}$$

From the spectral gap, for any centered f we have the Poincaré inequality

$$\int f^2\mathrm{d}\mu \leq \frac{1}{\lambda_1}\int |\nabla f|^2\mathrm{d}\mu. \tag{3.11}$$

Applying this inequality to each coordinate yields

$$\int |x|^2\mathrm{d}\mu \leq \frac{d}{\lambda_1}. \tag{3.12}$$

Therefore we have the upper bound

$$\sup_{x_0\in S}\mathrm{W}_2(X_t,\mu) \leq \mathrm{e}^{-\lambda_1 t}\Big(\sup_{x_0\in S}|x_0|^2 + \frac{d}{\lambda_1}\Big)^{1/2}. \tag{3.13}$$

Since $\sup_{x_0\in S}|x_0|^2 = c^2 d$, evaluating at $t_d = (1+\epsilon)\,t_*$ yields

$$\sup_{x_0\in S}\mathrm{W}_2(X_{t_d},\mu) \leq d^{-\epsilon/2}(c^2 + (\lambda_1)^{-1})^{1/2}. \tag{3.14}$$

Since $\liminf_{d\to\infty}\lambda_1 > 0$, letting d go to infinity concludes the proof.

Remark 3.4 (Weaker assumption)**.** *In the proof, the assumption* $\liminf_{d\to\infty}\lambda_1 > 0$ *could have been replaced by a slow enough growth of* $(\lambda_1)^{-1}$*, for instance* $(\lambda_1)^{-1} \leq (\log d)^\alpha$ *for some* $\alpha > 0$.

3.4. **Proof of Corollary 1.4.** Let us prove first the cutoffs in relative entropy and in total variation distance by using Corollary 1.3 on the Wasserstein distance. We drop the (d) superscript.

Upper bound. It is well known that $\mathrm{CD}(\rho,\infty)$ with $\rho>0$ implies that for all $t\geq t_0>0$,

$$\mathrm{H}(X_t\,|\,\mu)\leq \mathrm{e}^{-2\rho(t-t_0)}\mathrm{H}(X_{t_0}\,|\,\mu). \tag{3.15}$$

The role of $t_0>0$ is to benefit from the fact that the law of X_{t_0} is absolutely continuous with respect to μ, while the law of X_0 is a Dirac mass. On the other hand, following for instance [Bobkov et al., 2001, Lem. 4.2][3], $\mathrm{CD}(\rho,\infty)$ with $\rho>0$ implies[4] that for all $t>0$,

$$\mathrm{H}(X_t\,|\,\mu)\leq \frac{\rho\mathrm{e}^{-2\rho t}}{1-\mathrm{e}^{-2\rho t}}\mathrm{W}_2(X_0,\mu)^2\leq \frac{1}{2t}\mathrm{W}_2(X_0,\mu)^2. \tag{3.16}$$

Using (3.16) with $t=1$, and combining with (3.15), we obtain, for all $t\geq 1$,

$$2\mathrm{H}(X_t\,|\,\mu)\leq \mathrm{e}^{-2\rho(t-1)}\mathrm{W}_2(X_0,\mu)^2. \tag{3.17}$$

By combining with the general Csiszár–Kullback–Pinsker inequality

$$\mathrm{d}_{\mathrm{TV}}(\mu,\nu)^2\leq \frac{1}{2}\mathrm{H}(\nu\,|\,\mu), \tag{3.18}$$

we get finally, for all $t\geq 1$,

$$\mathrm{d}_{\mathrm{TV}}(X_t,\mu)^2\leq \frac{1}{2}\mathrm{H}(X_t\,|\,\mu)\leq \frac{1}{4}\mathrm{e}^{-2\rho(t-1)}\mathrm{W}_2(X_0,\mu)^2. \tag{3.19}$$

Recall that $\rho=\lambda_1$. Now, proceeding as in the proof of Corollary 1.3, we get, for all $c>0$,

$$\sup_{x_0\in B(m,c\sqrt{d})}\mathrm{W}_2(X_0,\mu)^2=\sup_{x_0\in B(m,c\sqrt{d})}\int|x-x_0|^2\mathrm{d}\mu(x)\leq (c^2+\lambda_1^{-1})d. \tag{3.20}$$

Using the definition of t_* and the assumption $\lambda_1=o_{d\to\infty}(\log(d))$, we get, for all $c,\varepsilon>0$,

$$\sup_{x_0\in B(m,c\sqrt{d})}\mathrm{e}^{-2\rho((1+\varepsilon)t_*-1)}\mathrm{W}_2(X_0,\mu)^2\leq \mathrm{e}^{O(\lambda_1)}d^{-\varepsilon}\xrightarrow[d\to\infty]{}0. \tag{3.21}$$

Therefore, going back to H and d_{TV}, we get, for all $c,\varepsilon>0$,

$$\lim_{d\to\infty}\sup_{x_0\in B(m,c\sqrt{d})}\mathrm{H}(X_{(1+\epsilon)t_*}\,|\,\mu)=0\text{ and }\lim_{d\to\infty}\sup_{x_0\in B(m,c\sqrt{d})}\mathrm{d}_{\mathrm{TV}}(X_{(1+\epsilon)t_*},\mu)=0. \tag{3.22}$$

The approach differs from the one in Boursier et al. [2023], in the way we regularize in (3.16) as well as in the way we control relative entropy, here via the Wasserstein distance.

Lower bound. Both total variation distance and relative entropy decrease by mappings,

$$\mathrm{d}_{\mathrm{TV}}(\mu\circ T^{-1},\nu\circ T^{-1})\leq \mathrm{d}_{\mathrm{TV}}(\mu,\nu)\quad\text{and}\quad \mathrm{H}(\nu\circ T^{-1}\,|\,\mu\circ T^{-1})\leq \mathrm{H}(\nu\,|\,\mu). \tag{3.23}$$

This contractivity argument is also at the heart of the lower bounds in Boursier et al. [2023]. It follows that we can bound from below the relative entropy and total variation mixing times by those of a suitable OU process, by taking a multi-eigenfunction map. The lower bounds for OU processes have been established for example in Lachaud [2005], Barrera et al. [2006], and [Boursier et al., 2023, Th. 1.2]. Therefore, in the setting of Corollary 1.4, for all $c>0$, $\varepsilon\in(0,1)$,

$$\lim_{d\to\infty}\sup_{x_0\in B(m,c\sqrt{d})}\mathrm{d}_{\mathrm{TV}}(X_{(1-\epsilon)t_*},\mu)=1\text{ and }\lim_{d\to\infty}\sup_{x_0\in B(m,c\sqrt{d})}\mathrm{H}(X_{(1-\epsilon)t_*}\,|\,\mu)=\infty, \tag{3.24}$$

when $X_0=x_0$, and for an arbitrary constant $c>0$. This is compatible with (3.18).

Another viewpoint for relative entropy is to get the Talagrand inequality

$$\mathrm{W}_2(X_t,\mu)^2\leq \frac{2}{\rho}\mathrm{H}(X_t\,|\,\mu) \tag{3.25}$$

from $\mathrm{CD}(\rho,\infty)$, and deduce the relative entropy lower bound from the W_2 lower bound.

[3] In Bobkov et al. [2001], it is at the heart of a derivation of the Otto–Villani HWI inequality, which contains the logarithmic Sobolev inequality as well as the Talagrand transportation inequality.

[4] The first inequality is valid for all $\rho\in\mathbb{R}$.

Let us prove the cutoff for the Fisher information. Following Bakry and Émery [1985], see also Bakry et al. [2014], $\mathrm{CD}(\rho,\infty)$ gives the logarithmic Sobolev inequality

$$\mathrm{H}(\nu \mid \mu) \le \frac{1}{2\rho}\mathrm{I}(\nu \mid \mu). \tag{3.26}$$

Taking $\nu = \mathrm{Law}(X_t)$ and using the cutoff lower bound for H give, for all $\varepsilon \in (0,1)$,

$$\lim_{d\to\infty} \sup_{x_0\in S} \mathrm{I}(X_{t_d} \mid \mu) = +\infty \quad \text{where} \quad t_d = (1-\varepsilon)\, t_*. \tag{3.27}$$

For an upper bound, we can use another side of the Bakry–Émery theorem which states that $\mathrm{CD}(\rho,\infty)$ gives the monotonicities

$$\partial_t \mathrm{H}(\mu_t \mid \mu) = -\mathrm{I}(\mu_t \mid \mu) \le 0 \quad \text{and} \quad \partial_t \mathrm{I}(\mu_t \mid \mu) \le -2\rho \mathrm{I}(\mu_t \mid \mu) \le 0 \tag{3.28}$$

where μ_t and μ stand for $\mathrm{Law}(X_t)$ and μ. The Grönwall lemma implies then the exponential decay of Fisher information: for all $t \ge t_1 > 0$,

$$\mathrm{I}(\mu_t \mid \mu) \le \mathrm{e}^{-2\rho(t-t_1)} \mathrm{I}(\mu_{t_1} \mid \mu) \tag{3.29}$$

(recall that μ_0 is a Dirac mass). To upper bound $\mathrm{I}(\mu_{t_1} \mid \mu)$, we write, for all $0 < t_0 < t_1$,

$$\mathrm{H}(\mu_{t_0} \mid \mu) - \mathrm{H}(\mu_{t_1} \mid \mu) = \int_{t_0}^{t_1} \mathrm{I}(\mu_s \mid \mu)\mathrm{d}s \ge (t_1 - t_0)\mathrm{I}(\mu_{t_1} \mid \mu) \tag{3.30}$$

where we have used the monotonicity of I. This gives, combined with (3.17), when $t_0 > 1$,

$$\mathrm{I}(\mu_{t_1} \mid \mu) \le \frac{\mathrm{H}(\mu_{t_0} \mid \mu)}{t_1 - t_0} \le \frac{\mathrm{e}^{-2\rho(t_0-1)}}{2(t_1-t_0)} \mathrm{W}_2(X_0,\mu)^2. \tag{3.31}$$

Finally, by using this inequality with $t_0 = t_d - 1$ and $t_1 = t_d$ where $t_d = (1+\varepsilon)\, t_*$ for an arbitrary fixed $\varepsilon > 0$, using (3.20), and $\rho = \lambda_1 = o_{d\to\infty}(\log(d))$, we get

$$\lim_{d\to\infty} \sup_{x_0\in S} \mathrm{I}(X_{t_d} \mid \mu) = 0. \tag{3.32}$$

Remark 3.5 (Weaker assumption)**.** *In the proof, the assumption $\lambda_1 = o(\log(d))$ could have been replaced by $\lambda_1 = d^{o(1)}$. This can be done by replacing, in the regularization step, the time 1 by time $\asymp 1/\lambda_1$, which produces a prefactor λ_1 for* H*, hence $\sqrt{\lambda_1}$ for $\mathrm{d_{TV}}$, and λ_1^2 for* I*.*

3.5. **Proof of Theorem 1.5.** Let $h(x) = x_1 + \cdots + x_d$ be the first symmetric Hermite polynomial. Its gradient is the constant vector $\nabla h = (1,\dots,1)$. Since $\mathrm{Hess}(V) = \rho I + \mathrm{Hess}(W) \ge \rho I$ as quadratic forms, we have $\lambda_1 \ge \rho$. Hence, to get $\lambda_1 = \rho$, it suffices to show that h is an eigenfunction of $-\mathscr{L}$ with eigenvalue ρ, and then apply the splitting theorem. The generator is

$$\mathscr{L} f = \Delta f - \rho x \cdot \nabla f - \nabla W \cdot \nabla f. \tag{3.33}$$

Since W is invariant along $\mathbb{R}(1,\dots,1)$, we have $\nabla W \cdot (1,\dots,1) = 0$, therefore,

$$\mathscr{L} h = 0 - \rho h - \nabla W \cdot (1,\dots,1) = -\rho h. \tag{3.34}$$

This gives $h \in E_1$ and $\lambda_1 = \rho$. Note that by rigidity, all the elements of E_1 are affine.

3.6. **Proof of Corollary 1.6.** We drop the (d) superscript. We have $\sum_{i=1}^d m_i = \int \sum_{i=1}^d x_i \mathrm{d}\mu(x) = 0$ since the image of μ by $(x_1,\dots,x_d) \mapsto x_1 + \cdots + x_d$ is a centered Gaussian.

Lower bound when $t_d = (1-\epsilon)\, t_*$. As per Theorem 1.5, $x_1 + \cdots + x_d$ is an eigenfunction. Since its squared $L^2(\mu)$ norm is $d\rho^{-1}$, $f_d(x) = \sqrt{\rho/d}(x_1 + \cdots + x_d)$ is a normalized eigenfunction. Now, for both $S = B(m, c\sqrt{d})$ and $S = m + [-c,c]^d$ we have

$$\sup_{x\in S} |f_d(x)| = c\sqrt{\rho}\sqrt{d}. \tag{3.35}$$

Hence the lower bound in Theorem 1.1 yields

$$\sup_{x_0\in S} \mathrm{W}_2(X_{t_d},\mu) \ge \mathrm{e}^{-\rho t_d} c\sqrt{d} = c d^{\epsilon/2} \xrightarrow[d\to\infty]{} +\infty. \tag{3.36}$$

Upper bound when $t_d = (1+\epsilon)\,t_*$. As in the proof of Corollary 1.3, the uniform convexity yields, via the Poincaré inequality, for all $1 \le i \le d$,

$$\int (x_i - m_i)^2 \mathrm{d}\mu(x) \le \rho^{-1}. \tag{3.37}$$

Hence when applying Theorem 1.1 we get

$$\sup_{x_0\in S} \mathrm{W}_2(X_{t_d},\mu) \le \mathrm{e}^{-\lambda_1 t_d}\Big(\sup_{x_0\in S}|x_0-m|^2 + d/\rho\Big)^{1/2} \le (c^2+\rho^{-1})^{1/2} d^{-\epsilon/2} \xrightarrow[d\to\infty]{} 0. \tag{3.38}$$

It remains to proceed as for Corollary 1.3 and Corollary 1.4 to conclude the proof.

3.7. **Proof of Theorem 1.7.** Here again, we drop the (d) superscript to simplify the notation. The spectral gap λ_1 of X satisfies $\lambda_1 \ge \kappa$, thus

$$\lim_{d\to\infty} \lambda_1 t_0 = +\infty, \tag{3.39}$$

which is the product condition in [Salez, 2025, Cor. 1], hence the cutoff in total variation distance. It remains to prove cutoff for the other cases. Let us start with the relative entropy lower bound. The Csiszár–Kullback–Pinsker inequality (3.18) gives

$$\liminf_{d\to\infty} \sup_{x_0\in S} \mathrm{H}(X_{(1-\varepsilon/2)t_0} \mid \mu) \ge 2\eta^2. \tag{3.40}$$

Moreover, from the exponential decay of entropy, for all $t>0$ and $t' \ge t$,

$$\mathrm{H}(X_{t'} \mid \mu) \le \mathrm{e}^{-2\kappa(t'-t)} \mathrm{H}(X_t \mid \mu). \tag{3.41}$$

Using $t' = (1-\frac{\varepsilon}{2})\,t_0$, $t = (1-\varepsilon)\,t_0$, and $\lim_{d\to\infty} \kappa t_0 = +\infty$, we get

$$\liminf_{d\to\infty} \sup_{x_0\in S} \mathrm{H}(X_{(1-\varepsilon)t_0} \mid \mu) \ge \mathrm{e}^{\varepsilon \limsup_{d\to\infty} \kappa t_0} 2\eta^2 = +\infty. \tag{3.42}$$

For the upper bound, from [Salez, 2025, Proof of Th. 1], we get

$$\limsup_{d\to\infty} \sup_{x_0\in S} \mathrm{H}\big(X_{(1+\frac{\varepsilon}{2})t_0} \mid \mu\big) \le C_{\eta,\varepsilon} < \infty. \tag{3.43}$$

Using the exponential decay of the relative entropy and $\lim_{d\to\infty} \kappa t_0 = +\infty$, we get

$$\limsup_{d\to\infty} \sup_{x_0\in S} \mathrm{H}(X_{(1+\varepsilon)t_0} \mid \mu) \le \mathrm{e}^{-\varepsilon \liminf_{d\to\infty} \kappa t_0} C_{\eta,\varepsilon} = 0. \tag{3.44}$$

It remains to establish the cutoffs for W_2 and I. Let us recall the extra assumptions $\underline{\lim}_{d\to\infty} \kappa > 0$ and $\lim_{d\to\infty} t_0 = +\infty$.

Now, regarding W_2, the upper bound comes from the one for H via the Talagrand inequality $\kappa \mathrm{W}_2(X,\mu)^2 \le 2\mathrm{H}(X \mid \mu)$ and the assumption $\underline{\lim}_d \kappa > 0$, while the lower bound comes from the regularization (3.16) and the semigroup property that give, with $s = \frac{\varepsilon}{2} t_0$,

$$\mathrm{H}(X_{(1-\frac{\varepsilon}{2})t_0} \mid \mu) \le \frac{1}{2s} \mathrm{W}_2(X_{(1-\varepsilon)t_0},\mu)^2, \tag{3.45}$$

and then the lower bound for H and the assumption $t_0 \to \infty$:

$$\sup_{x_0\in S} \mathrm{W}_2(X_{(1-\varepsilon)t_0},\mu)^2 \ge 2s \times 2\eta^2 = 2\varepsilon\eta^2 t_0 \xrightarrow[d\to\infty]{} \infty. \tag{3.46}$$

Finally, for I, the lower bound comes from the one for H via the logarithmic Sobolev inequality $2\kappa \mathrm{H}(X \mid \mu) \le \mathrm{I}(X \mid \mu)$ and the assumption $\underline{\lim}_d \kappa > 0$, while the upper bound comes by regularization from (3.30) and the semigroup property to get

$$\mathrm{I}(X_{t_1} \mid \mu) \le \frac{\mathrm{H}(X_{t_0'} \mid \mu)}{t_1 - t_0'} \tag{3.47}$$

where $t_0' = (1+\frac{\varepsilon}{2})\,t_0$ and $t_1 = (1+\varepsilon)\,t_0$, then the upper bound on H and the assumption $t_0 \to \infty$:

$$\sup_{x_0\in S} \mathrm{I}(X_{(1+\varepsilon)t_0} \mid \mu) \le \frac{\sup_{x_0\in S} \mathrm{H}(X_{(1+\frac{\varepsilon}{2})t_0} \mid \mu)}{\frac{\varepsilon}{2} t_0} \xrightarrow[d\to\infty]{} 0. \tag{3.48}$$

3.8. **Proof of Theorem 1.9.** The proof is exactly the same as in the Euclidean setting (Theorem 1.1), up to the use of the Riemannian splitting (Theorem 3.6) below, and the fact that first eigenfunctions are affine through the splitting (Lemma 3.7). We shall hence only discuss these two elements, and omit the repetition of the proof. In this setting, the rigidity theorem of Cheng and Zhou [2017] when $\lambda_1 = \rho$ is the following.

Theorem 3.6 (Riemannian splitting with Gaussian factor)**.** *Let (M, g, μ) be a smooth, connected, complete weighted Riemannian manifold without boundary, with probability measure $\mu = \mathrm{e}^{-V}$. If for some $\rho > 0$,*

$$\mathrm{CD}(\rho, \infty) \text{ is satisfied and } \lambda_1 = \rho, \tag{3.49}$$

then (M, g, μ) is isometric to a product weighted Riemannian manifold

$$(\mathbb{R}^{k_1}, |\cdot|_2, \gamma_{k_1,\rho^{-1}}) \times (M', g', \mu') = (\mathbb{R}^{k_1} \times M', |\cdot|_2 \oplus g', \gamma_{k_1,\rho^{-1}} \otimes \mu') \tag{3.50}$$

where

- *k_1 is the dimension of the eigenspace of $-\mathscr{L}$ associated with the eigenvalue λ_1*
- *$\gamma_{k_1,\rho^{-1}}$ is the centered Gaussian law on $\mathbb{R}^{k_1}$ with covariance matrix $\rho^{-1}\mathrm{Id}_{k_1}$*
- *(M', g', μ') is a weighted Riemannian manifold satisfying $\mathrm{CD}(\rho, \infty)$.*

This was proved by Cheng and Zhou [2017] in the setting of complete smooth weighted manifolds, and by Gigli et al. [2020] in the more general setting of RCD spaces. Splitting theorems for manifolds satisfying a curvature constraint and optimizing certain geometric quantities is a well-studied problem in Riemannian geometry, going back to the Cheeger–Gromoll splitting theorem for non-negatively curved manifolds containing infinite geodesics.

We now state a structure lemma which enters the proof of Theorem 3.6.

Lemma 3.7 (Rigidity for eigenfunctions when $\lambda_1 = \rho$)**.** *Under the setting of Theorem 3.6, and through the isometry that it provides, any element of E_1 is of the form $a \cdot p(x)$ with p the projection on the Euclidean factor of dimension k_1, $a \in \mathbb{R}^{k_1}$, $|a| = \sqrt{\rho}\|f\|_{L^2(\mu)}$. Moreover, if $f_i = a_i \cdot p(x)$, for $i = 1, 2$, are two orthogonal eigenfunctions, then $a_1 \perp a_2$.*

Proof. What follows is a broad sketch, where we focus on justifying the form of the eigenfunctions, but do not discuss in too much detail the splitting of the space, that was established in Cheng and Zhou [2017]. See also Gigli et al. [2020] for a full proof in the non-smooth setting of RCD spaces. From the integrated Bochner formula, for any g in the domain of $\mathscr{L}$ we have

$$\int -\Gamma(g, \mathscr{L}g)\mathrm{d}\mu \geq \rho \int \Gamma(g)\mathrm{d}\mu + \int \|\mathrm{Hess}(g)\|_{\mathrm{HS}}^2 \mathrm{d}\mu. \tag{3.51}$$

Taking g to be an eigenfunction with eigenvalue $\lambda_1 = \rho$, we get

$$\rho \int \Gamma(f)\mathrm{d}\mu \geq \rho \int \Gamma(f)\mathrm{d}\mu + \int \|\mathrm{Hess}(f)\|_{\mathrm{HS}}^2 \mathrm{d}\mu \tag{3.52}$$

which forces $\mathrm{Hess}(f) = 0$ almost everywhere, and thus everywhere since eigenfunctions are smooth. Therefore f is affine, and it is non-constant whenever $f \neq 0$. For a nonzero eigenfunction f, ∇f is a non-trivial parallel vector field of constant norm. Completeness ensures that its unit normalization has a flow defined for all real times. Along this flow, f increases at the constant rate $|\nabla f|$, so every flow line meets $f^{-1}(0)$ exactly once. Since the field is parallel, its flow gives an isometry $\mathbb{R} \times f^{-1}(0) \to M$, along which f is linear in the first coordinate. Repeating this strategy for successive orthogonal eigenfunctions gives a splitting with a k_1-dimensional Euclidean factor. See Cheng and Zhou [2017] for full details. And once the splitting is established, we can view the eigenfunctions as functions on $\mathbb{R}^{k_1}$.

Since for an affine function $f = a \cdot x + b$, we have $\Gamma(f) = |a|^2$, and

$$|a|^2 = \int \Gamma(f)\mathrm{d}\mu = -\int f(\mathscr{L}f)\mathrm{d}\mu = \rho\|f\|_{L^2(\mu)}^2. \tag{3.53}$$

Let us now compute $\Gamma(f_1, f_2)$ where f_1, f_2 are orthogonal eigenfunctions. We have

$$\Gamma(f_1, f_2) = \frac{1}{4}(\Gamma(f_1 + f_2) - \Gamma(f_1 - f_2)) = \frac{\rho}{4}(\|f_1 + f_2\|^2_{L^2(\mu)} - \|f_1 - f_2\|^2_{L^2(\mu)}) = 0 \tag{3.54}$$

where we used the fact that $f_1 \pm f_2$ are also eigenfunctions, with the same eigenvalue λ_1. But since $\Gamma(f_1, f_2) = \langle a_1, a_2 \rangle$, the vectors driving f_1 and f_2 must be orthogonal. □

REFERENCES

D. Bakry and M. Émery. Diffusions hypercontractives. Sémin. de probabilités XIX, Univ. Strasbourg 1983/84, Proc., Lect. Notes Math. 1123, 177-206 (1985)., 1985.

D. Bakry and Z. Qian. Some new results on eigenvectors via dimension, diameter, and Ricci curvature. *Adv. Math.*, 155 (1):98–153, 2000. doi: 10.1006/aima.2000.1932.

D. Bakry, I. Gentil, and M. Ledoux. *Analysis and geometry of Markov diffusion operators*, volume 348 of *Grundlehren Math. Wiss.* Springer, 2014. doi: 10.1007/978-3-319-00227-9.

G. Barrera. Abrupt convergence for a family of Ornstein-Uhlenbeck processes. *Braz. J. Probab. Stat.*, 32(1):188–199, 2018. doi: 10.1214/16-BJPS337.

J. Barrera, B. Lachaud, and B. Ycart. Cut-off for n-tuples of exponentially converging processes. *Stochastic Process. Appl.*, 116(10):1433–1446, 2006. doi: 10.1016/j.spa.2006.03.003.

F. Barthe and B. Klartag. Spectral gaps, symmetries and log-concave perturbations. *Bull. Hell. Math. Soc.*, 64:1–31, 2020. URL `bulletin.math.uoc.gr/vol/64/64-1-31.pdf`.

J. Bertrand and M. Fathi. Stability of eigenvalues and observable diameter in RCD$(1,\infty)$ spaces. *J. Geom. Anal.*, 32(11): 38, 2022. doi: 10.1007/s12220-022-00999-9. Id/No 270.

S. G. Bobkov, I. Gentil, and M. Ledoux. Hypercontractivity of Hamilton-Jacobi equations. *J. Math. Pures Appl. (9)*, 80 (7):669–696, 2001. doi: 10.1016/S0021-7824(01)01208-9.

J. Boursier, D. Chafaï, and C. Labbé. Universal cutoff for Dyson Ornstein Uhlenbeck process. *Probab. Theory Relat. Fields*, 185(1-2):449–512, 2023. doi: 10.1007/s00440-022-01158-5.

D. Chafaï, 2024. `https://djalil.chafai.net/blog/2024/01/27/cutoff-for-markov-processes/`.

D. Chafaï and J. Lehec. On Poincaré and logarithmic Sobolev inequalities for a class of singular Gibbs measures. In *Geometric aspects of functional analysis. Vol. I*, volume 2256 of *Lecture Notes in Math.*, pages 219–246. Springer, 2020. doi: 10.1007/978-3-030-36020-7_10.

G.-Y. Chen and L. Saloff-Coste. The cutoff phenomenon for ergodic Markov processes. *Electron. J. Probab.*, 13:no. 3, 26–78, 2008. doi: 10.1214/EJP.v13-474.

X. Chen, X.-M. Li, and B. Wu. Logarithmic heat kernel estimates without curvature restrictions. *Ann. Probab.*, 51(2): 442–477, 2023. doi: 10.1214/22-aop1599.

X. Cheng and D. Zhou. Eigenvalues of the drifted Laplacian on complete metric measure spaces. *Commun. Contemp. Math.*, 19(1):17, 2017. doi: 10.1142/S0219199716500012. Id/No 1650001.

T. A. Courtade and M. Fathi. Stability of the Bakry-émery theorem on $\mathbb{R}^n$. *J. Funct. Anal.*, 279(2):28, 2020. doi: 10.1016/j.jfa.2020.108523. Id/No 108523.

G. De Philippis and A. Figalli. Rigidity and stability of Caffarelli's log-concave perturbation theorem. *Nonlinear Anal., Theory Methods Appl., Ser. A, Theory Methods*, 154:59–70, 2017. doi: 10.1016/j.na.2016.10.006.

A. Engoulatov. A universal bound on the gradient of logarithm of the heat kernel for manifolds with bounded Ricci curvature. *J. Funct. Anal.*, 238(2):518–529, 2006. doi: 10.1016/j.jfa.2006.02.013.

L. C. Evans. *Partial Differential Equations*, volume 19 of *Graduate Studies in Mathematics*. American Mathematical Society, Providence, RI, 2 edition, 2010.

M. Fathi, I. Gentil, and J. Serres. Stability estimates for the sharp spectral gap bound under a curvature-dimension condition. *Ann. Inst. Fourier*, 74(6):2425–2459, 2024. doi: 10.5802/aif.3608.

N. Gigli, A. Mondino, and G. Savaré. Convergence of pointed non-compact metric measure spaces and stability of Ricci curvature bounds and heat flows. *Proc. Lond. Math. Soc. (3)*, 111(5):1071–1129, 2015. doi: 10.1112/plms/pdv047.

N. Gigli, C. Ketterer, K. Kuwada, and S.-I. Ohta. Rigidity for the spectral gap on RCD(K,∞)-spaces. *Am. J. Math.*, 142 (5):1559–1594, 2020. doi: 10.1353/ajm.2020.0039.

A. Grigor'yan. *Heat kernel and analysis on manifolds*, volume 47 of *AMS/IP Stud. Adv. Math.* American Mathematical Society (AMS), 2009.

E. P. Hsu. *Stochastic analysis on manifolds*, volume 38 of *Grad. Stud. Math.* American Mathematical Society, 2002.

B. Klartag. Logarithmic bounds for isoperimetry and slices of convex sets. *Ars Inven. Anal.*, pages Paper No. 4, 17, 2023. ISSN 2769-8505.

B. Lachaud. Cut-off and hitting times of a sample of Ornstein-Uhlenbeck processes and its average. *J. Appl. Probab.*, 42(4):1069–1080, 2005. doi: 10.1239/jap/1134587817.

M. Lassalle. Polynômes de Hermite généralisés. *C. R. Acad. Sci. Paris Sér. I Math.*, 313(9):579–582, 1991.

A. Lichnérowicz. *Géométrie des groupes de transformations*, volume 3 of *Trav. Rech. Math.* Dunod, Paris, 1958.

C. H. Mai and S.-I. Ohta. Quantitative estimates for the Bakry-Ledoux isoperimetric inequality. *Comment. Math. Helv.*, 96(4):693–739, 2021. doi: 10.4171/CMH/523.

P.-L. Méliot. The cut-off phenomenon for Brownian motions on compact symmetric spaces. *Potential Anal.*, 40(4): 427–509, 2014. doi: 10.1007/s11118-013-9356-7.

M. Obata. Certain conditions for a Riemannian manifold to be isometric with a sphere. *J. Math. Soc. Japan*, 14:333–340, 1962. doi: 10.2969/jmsj/01430333.

Y. Ollivier. A visual introduction to Riemannian curvatures and some discrete generalizations. In *Analysis and geometry of metric measure spaces. Lecture notes of the 50th Séminaire de Mathématiques Supérieures (SMS), Montréal, Canada, June 27 – July 8, 2011*, pages 197–220. American Mathematical Society, 2013.

F. Pedrotti and J. Salez. A new cutoff criterion for non-negatively curved chains. preprint arXiv:2501.13079v1, 2025.

O. S. Rothaus. Diffusion on compact Riemannian manifolds and logarithmic Sobolev inequalities. *J. Functional Analysis*, 42(1):102–109, 1981. doi: 10.1016/0022-1236(81)90049-5.

J. Salez. Cutoff for non-negatively curved Markov chains. *J. Eur. Math. Soc. (JEMS)*, 26(11):4375–4392, 2024. doi: 10.4171/JEMS/1348.

J. Salez. Cutoff for non-negatively curved diffusions. preprint arXiv:2501.01304v1, 2025.

L. Saloff-Coste. Precise estimates on the rate at which certain diffusions tend to equilibrium. *Math. Z.*, 217(4):641–677, 1994. doi: 10.1007/BF02571965.

L. Saloff-Coste. Lectures on finite Markov chains. In *Saint-Flour summer school 1996*, volume 1665 of *Lecture Notes in Math.*, pages 301–413. Springer, 1997. doi: 10.1007/BFb0092621.

F. Santambrogio. *Optimal transport for applied mathematicians*, volume 87 of *Progress in Nonlinear Differential Equations and their Applications*. Birkhäuser/Springer, 2015. doi: 10.1007/978-3-319-20828-2. Calculus of variations, PDEs, and modeling.

M.-K. von Renesse and K.-T. Sturm. Transport inequalities, gradient estimates, entropy and Ricci curvature. *Commun. Pure Appl. Math.*, 58(7):923–940, 2005. doi: 10.1002/cpa.20060.

(DC) DMA, ÉCOLE NORMALE SUPÉRIEURE, 45 RUE D'ULM, 75230, PARIS; AND CEREMADE, UNIVERSITÉ PARIS-DAUPHINE, PLACE DU MARÉCHAL DE LATTRE DE TASSIGNY, 75775, PARIS; PSL, CNRS

Email address: djalil@chafai.net

URL: https://djalil.chafai.net/

(MF) UNIVERSITÉ PARIS CITÉ AND SORBONNE UNIVERSITÉ, CNRS, LABORATOIRE JACQUES-LOUIS LIONS AND LABORATOIRE DE PROBABILITÉS, STATISTIQUE ET MODÉLISATION, F-75013 PARIS, FRANCE, AND DMA, ÉCOLE NORMALE SUPÉRIEURE, UNIVERSITÉ PSL, CNRS, 75005 PARIS, FRANCE, AND INSTITUT UNIVERSITAIRE DE FRANCE

Email address: mfathi@lpsm.paris